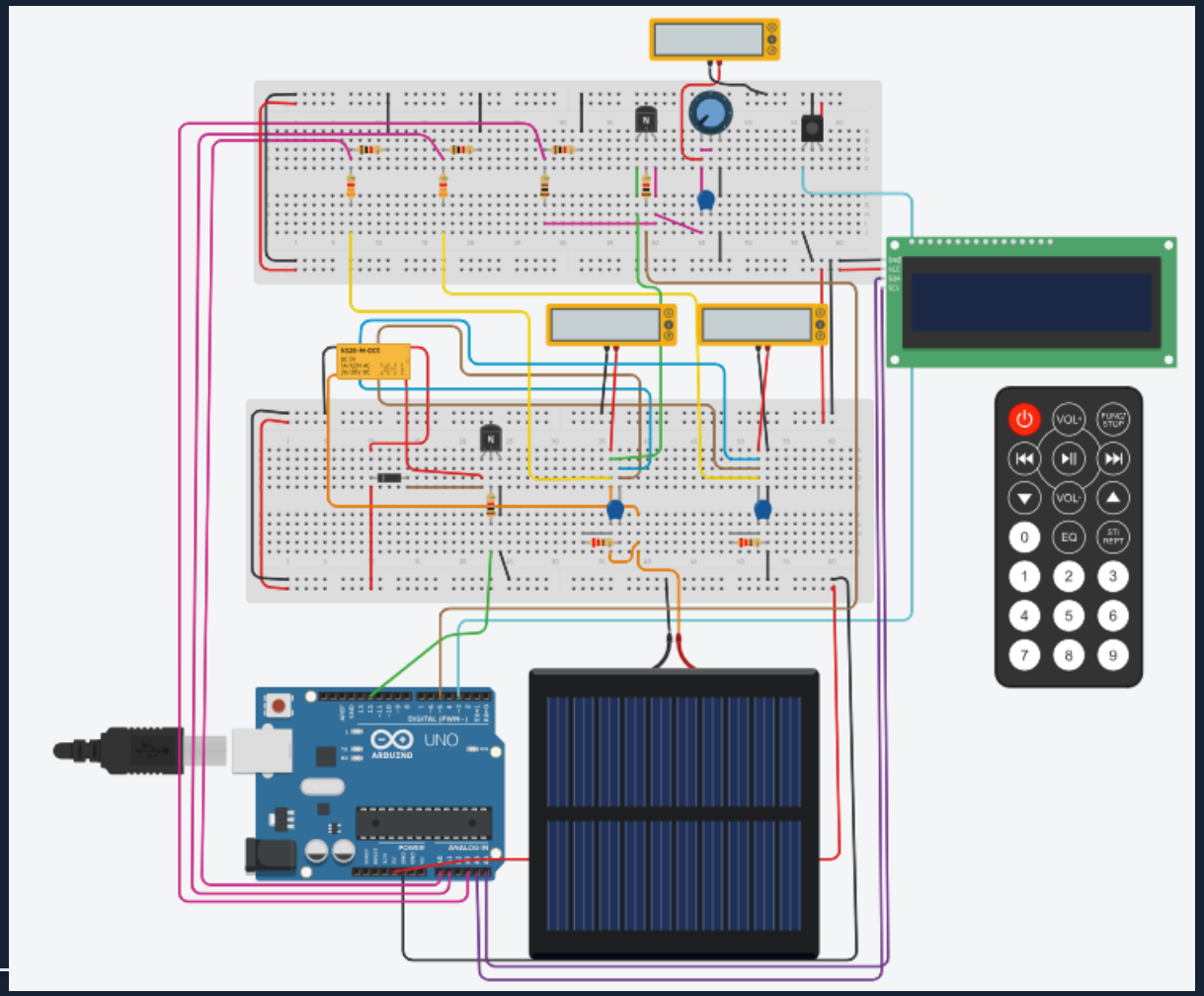
Replacing Batteries with a Substitute in a Solar System

Dynamic Series-Parallel Capacitor Switching & Closed-Loop Regulation



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Background & Objectives

Problem

Solar energy needs storage when input is unavailable or insufficient. Conventional batteries have chemical and maintenance limitations.

Proposed Solution

Use a capacitor bank as the storage element and dynamically configure it as series or parallel.

- Store energy using capacitors.
- Switch dynamically between parallel and series configurations.
- Use parallel operation when the required voltage is already available.
- Use series operation when additional voltage headroom is required.
- Provide regulated output voltage selected by the user through an IR remote.

Core System Architecture & Components

Solar panel/cell

Provides the input energy.

Relay

Changes capacitor topology between series and parallel.

Capacitor bank

Acts as the battery substitute and stores energy.

NPN transistor

Variable control element for output regulation.

Voltage dividers

Scale system voltages to safe Arduino input levels.

IR receiver + LCD

User input, feedback and system-state display.



System Block Diagram

- Solar panel / energy source
- Capacitor bank
- Series/parallel relay switching
- NPN transistor and output
- Arduino control unit
- Voltage sensors A0, A1 and A3
- IR receiver on Pin 3
- PWM control on Pin 5
- LCD display

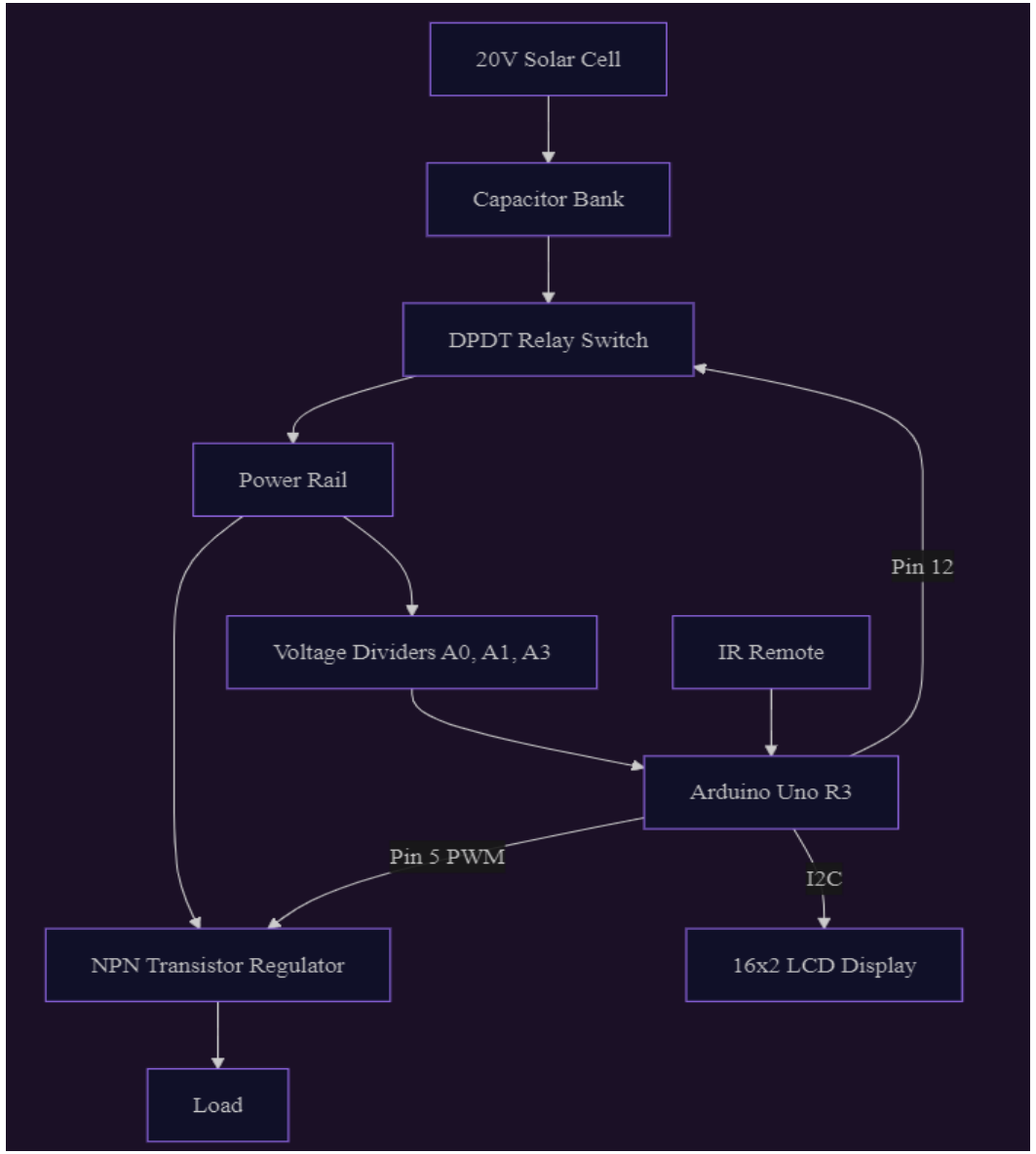


Diagram 1 – Voltage Sensing and Display

Shows the voltage-divider resistor network, Arduino connections and LCD interface.

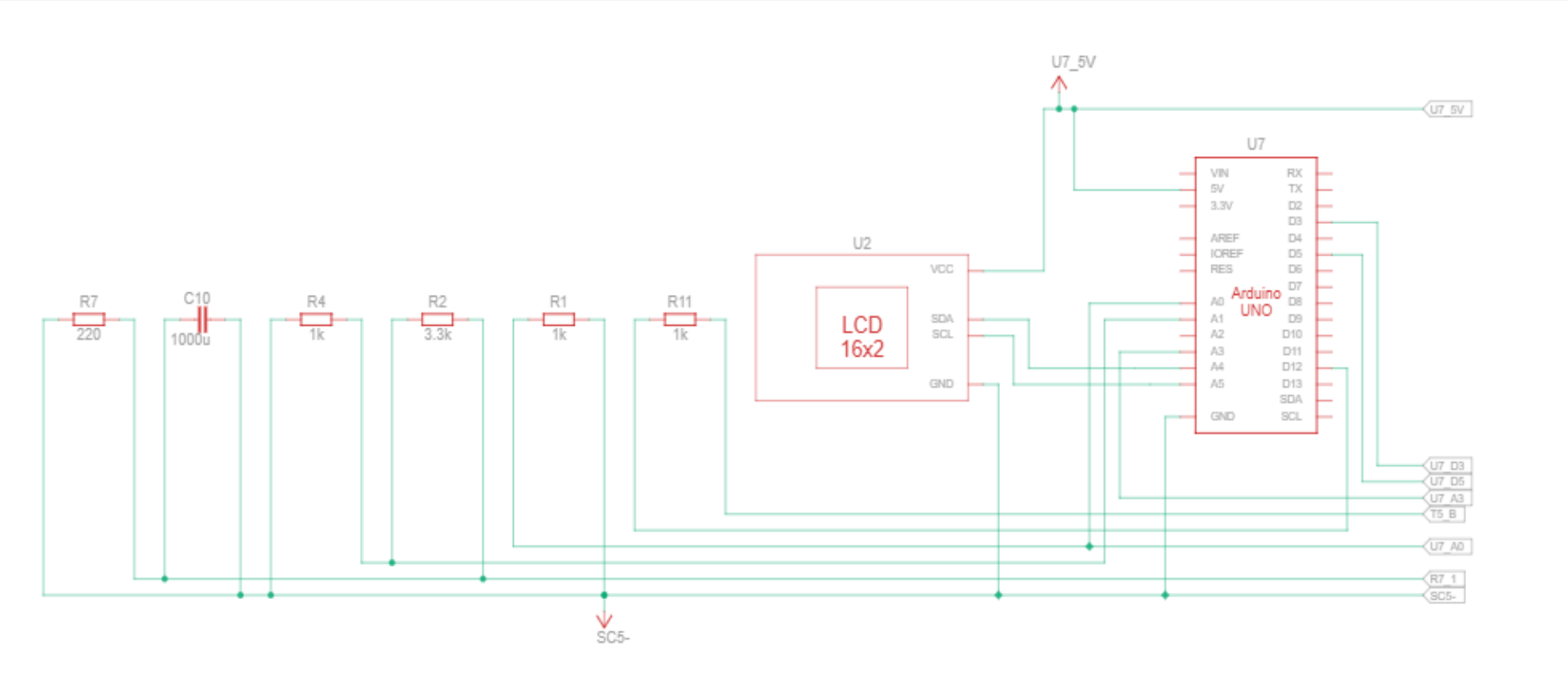


Diagram 2 – Capacitor Storage and Switching

Shows the relay, capacitors, and associated power and protection connections.

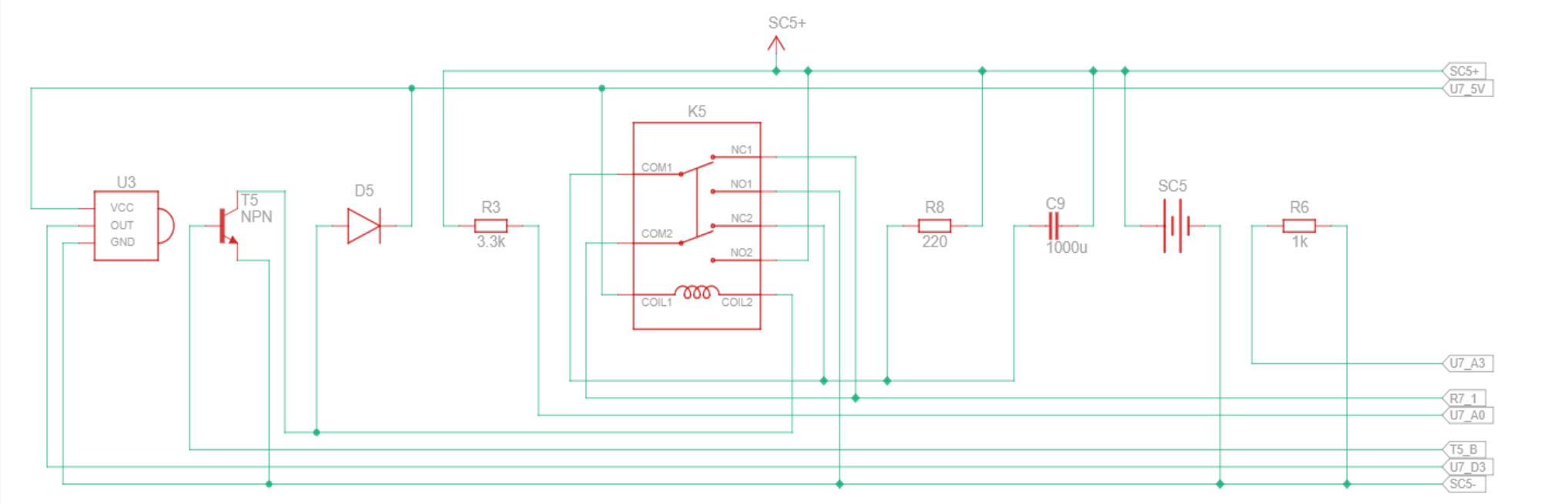
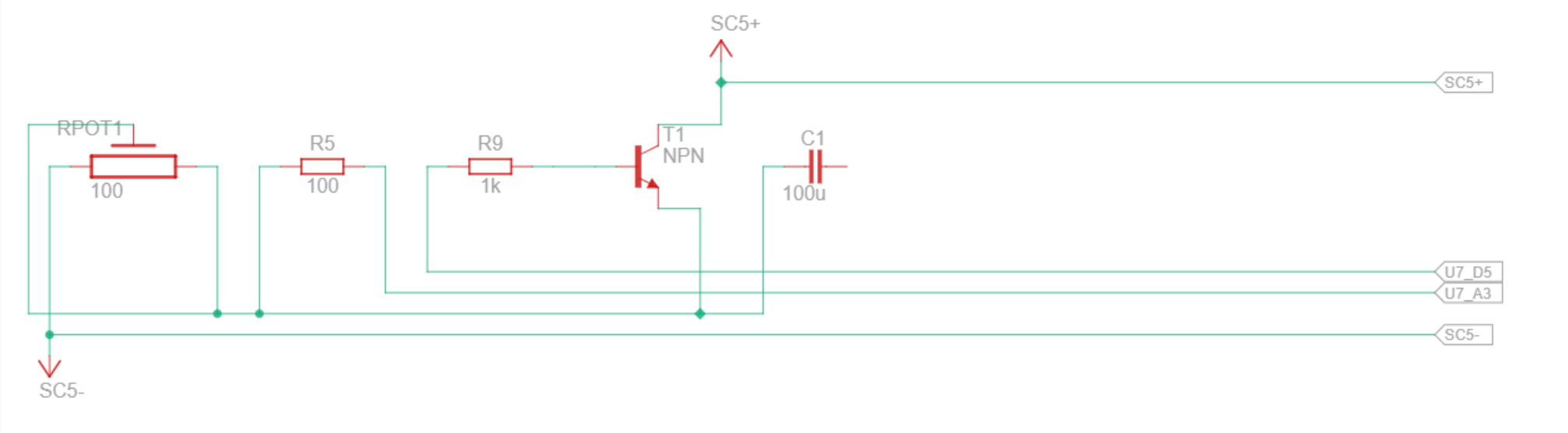


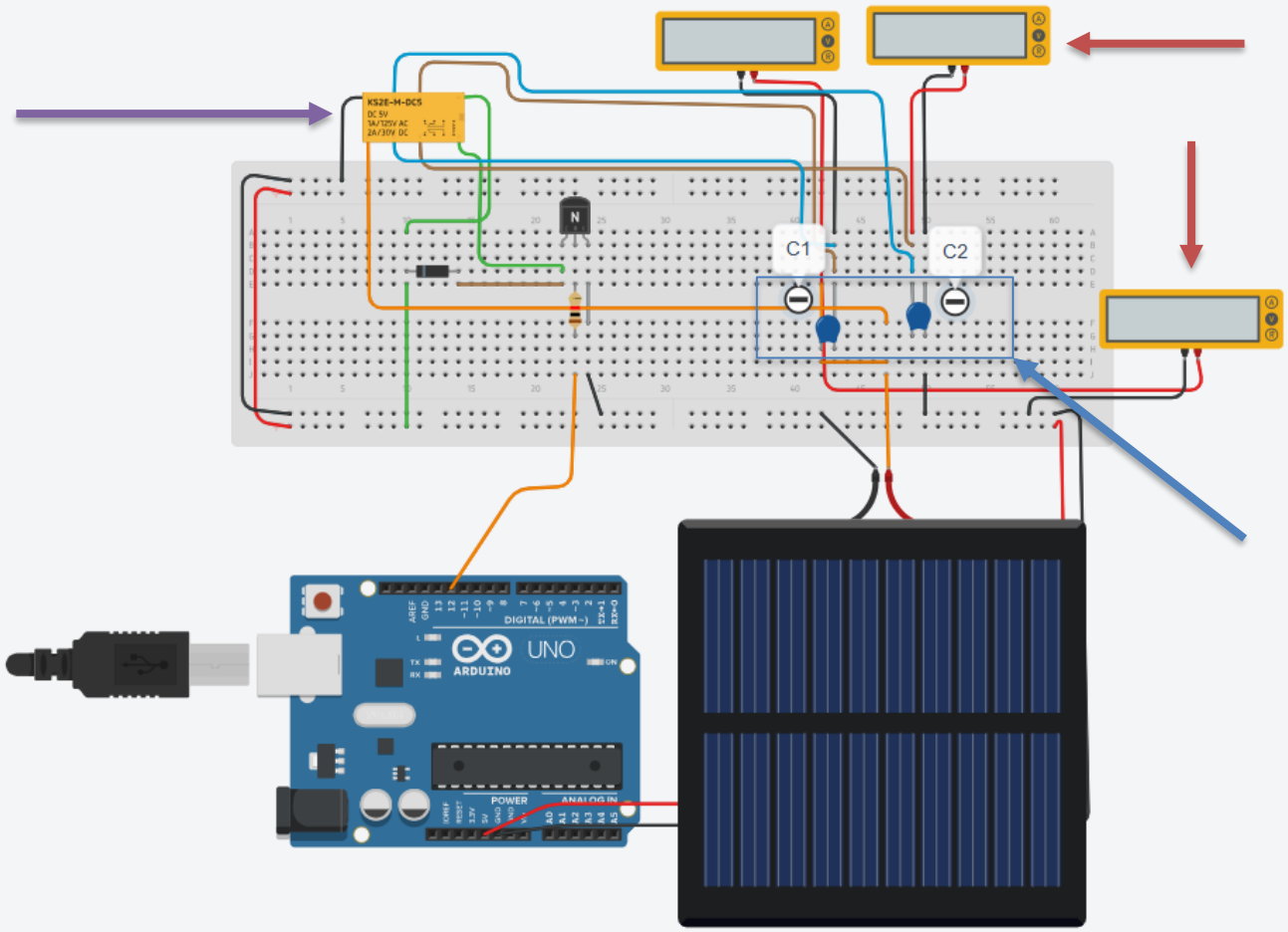
Diagram 3 – Transistor Control Stage

Shows the additional transistor, resistor and capacitor connections associated with the control stage.



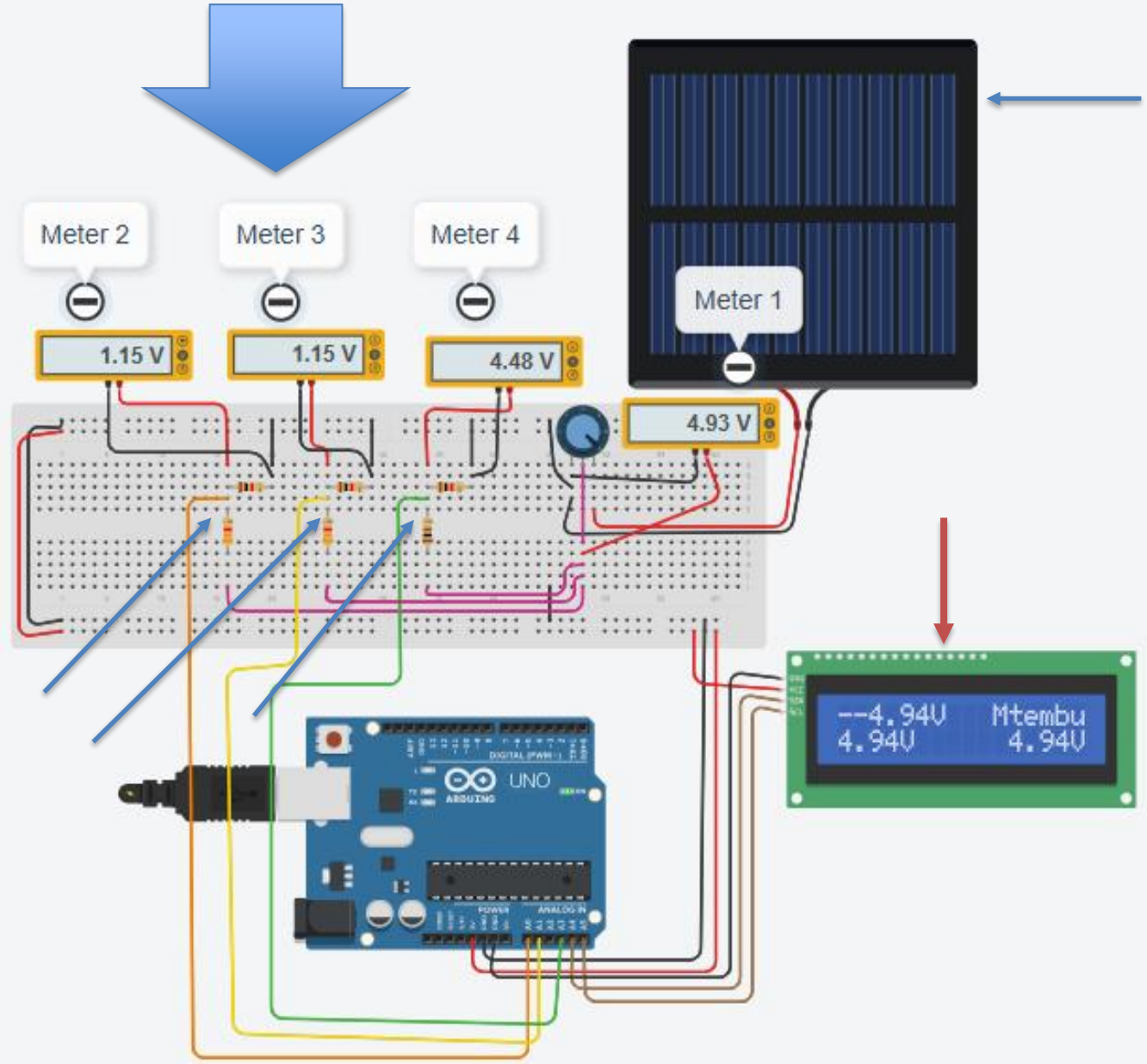
Phase 1 — Base Power & Topology Switching

- Purpose: establish the basic energy-storage and switching circuit.
- Components: solar cell, two capacitors, relay and protection components.
- Parallel: capacitors share the electrical path and operate at the same voltage.
- Series: capacitor voltages combine to provide a higher system voltage.
- Relay physically changes the electrical connection between the capacitors.
- Initial verification used a serial monitor and physical voltmeter.



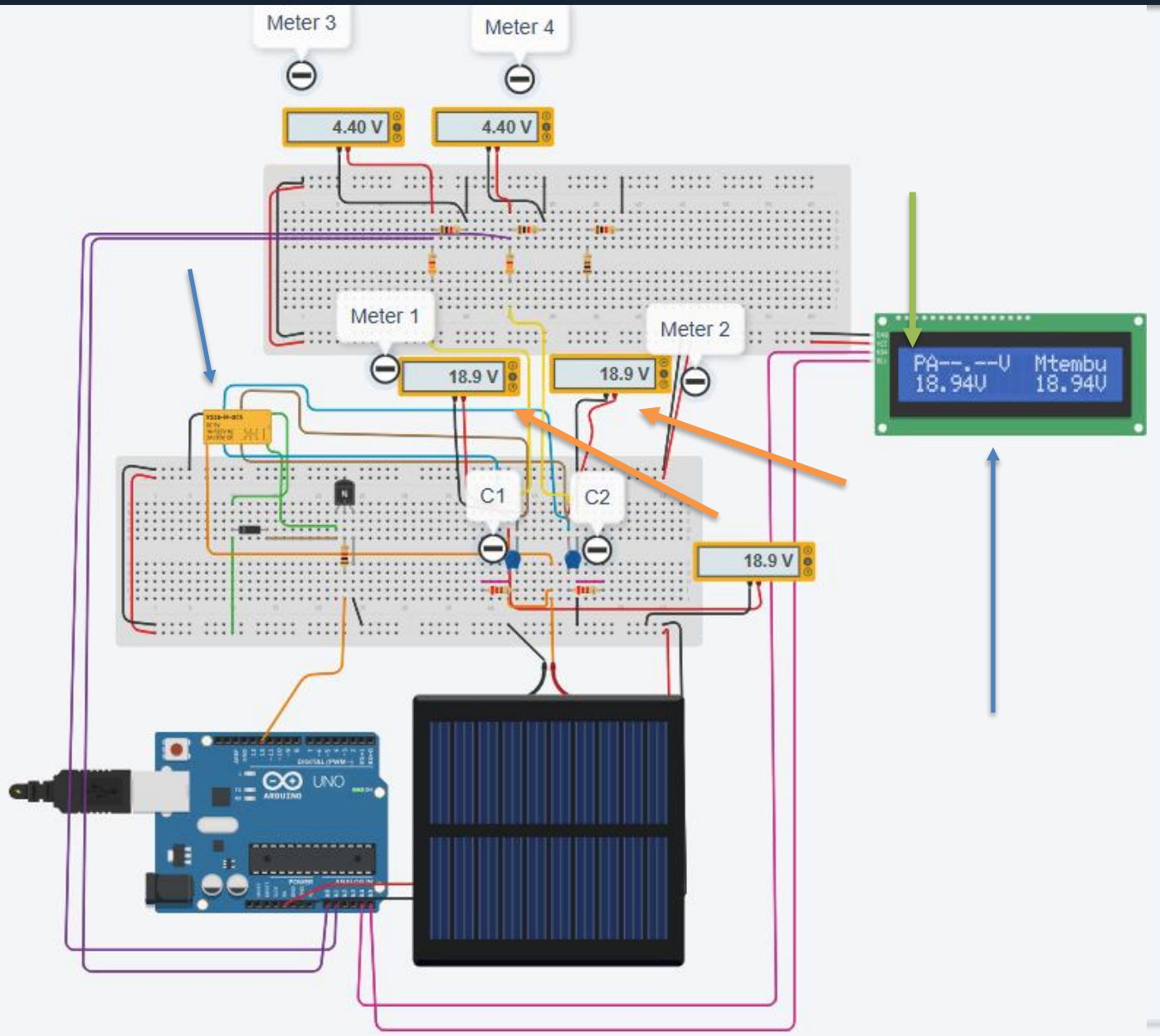
Phase 2 — Safe Telemetry & Calibration

- Purpose: safely measure system voltages that can reach approximately 20 V.
- Three voltage-divider circuits are used on A0, A1 and A3.
- Dividers reduce the voltage to a safe range for Arduino analogue inputs.
- Software converts ADC readings back to calibrated system-voltage values.
- LCD displays the calibrated measurements for live testing.
- Calibration values were checked against a physical voltmeter.



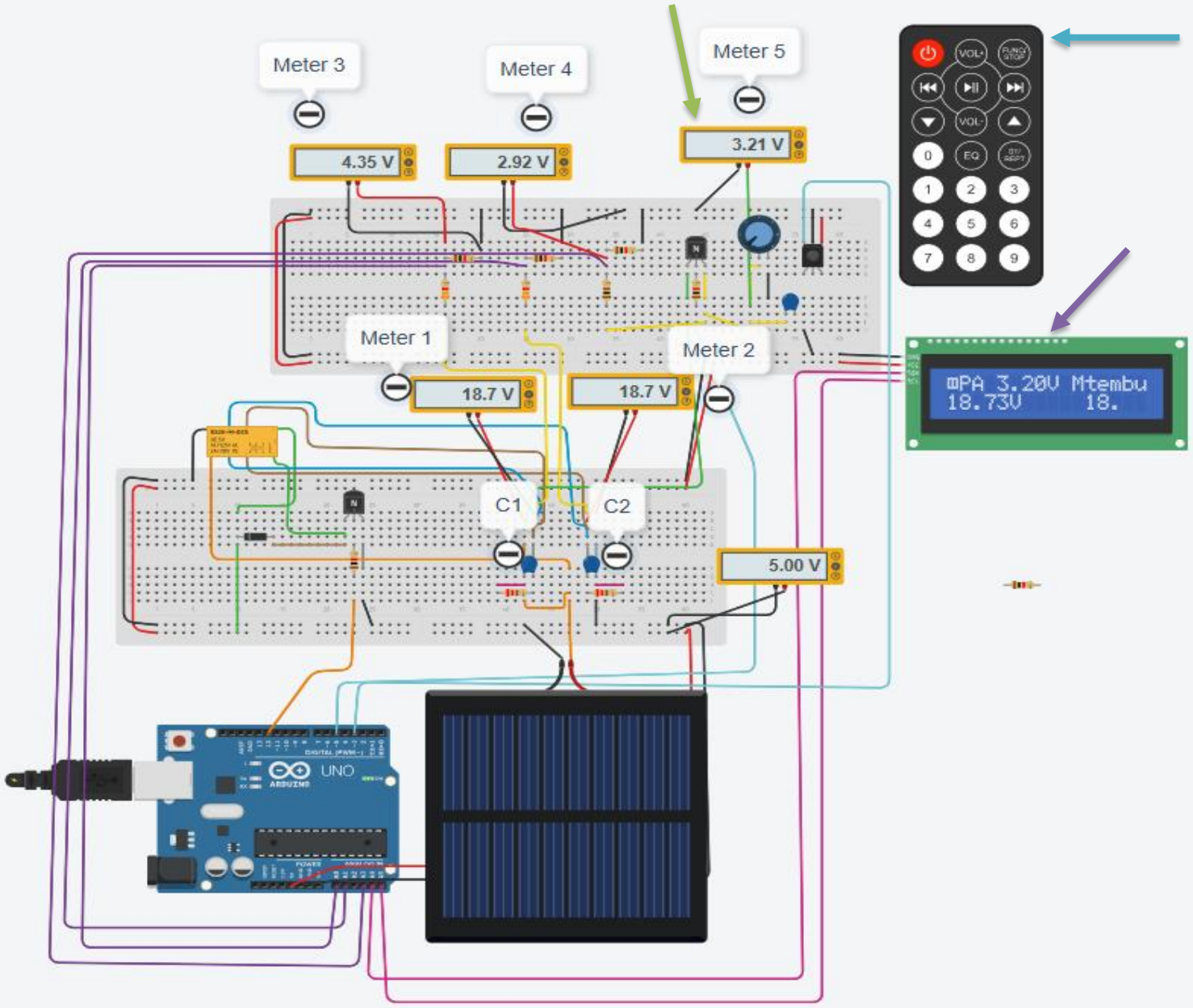
Phase 3 — Telemetry Integration

- Purpose: merge relay switching with live voltage sensing.
- Arduino reads the capacitor/system voltages while controlling the relay.
- LCD displays the measured values in real time.
- Voltage readings change as the relay moves between series and parallel.
- Telemetry provides immediate confirmation of the electrical state.
- Result: switching and measurement operate as one subsystem.



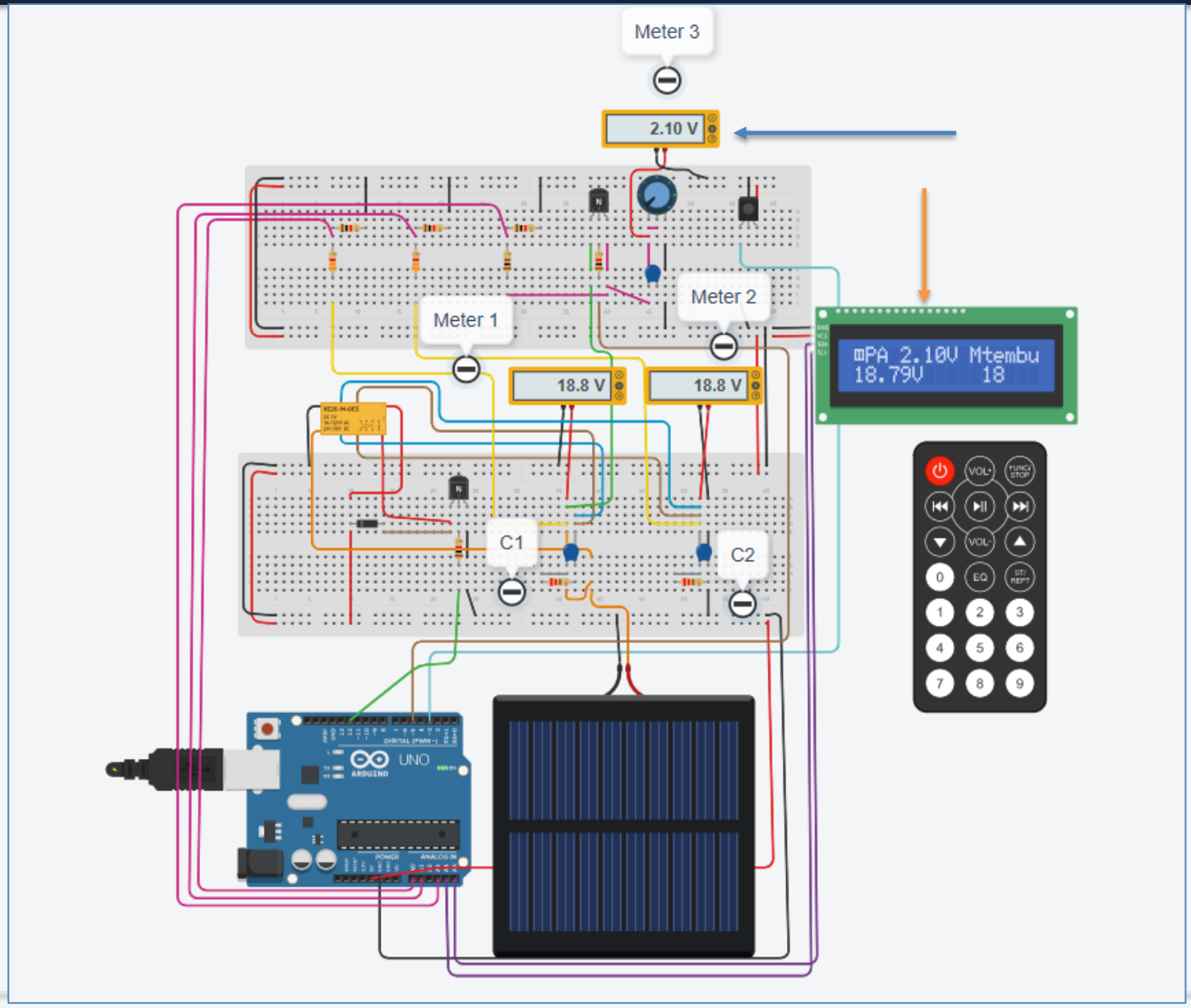
Phase 4 — IR Remote Input Capture

- Purpose: allow the user to select the required regulated output voltage.
- IR receiver is connected to Arduino Pin 3.
- Software decodes the remote command and identifies the target voltage.
- LCD displays the selected value as: Input = 3.20V
- Target voltage is displayed for approximately 1 second and then stored.
- Result: output target can be changed without editing the program.



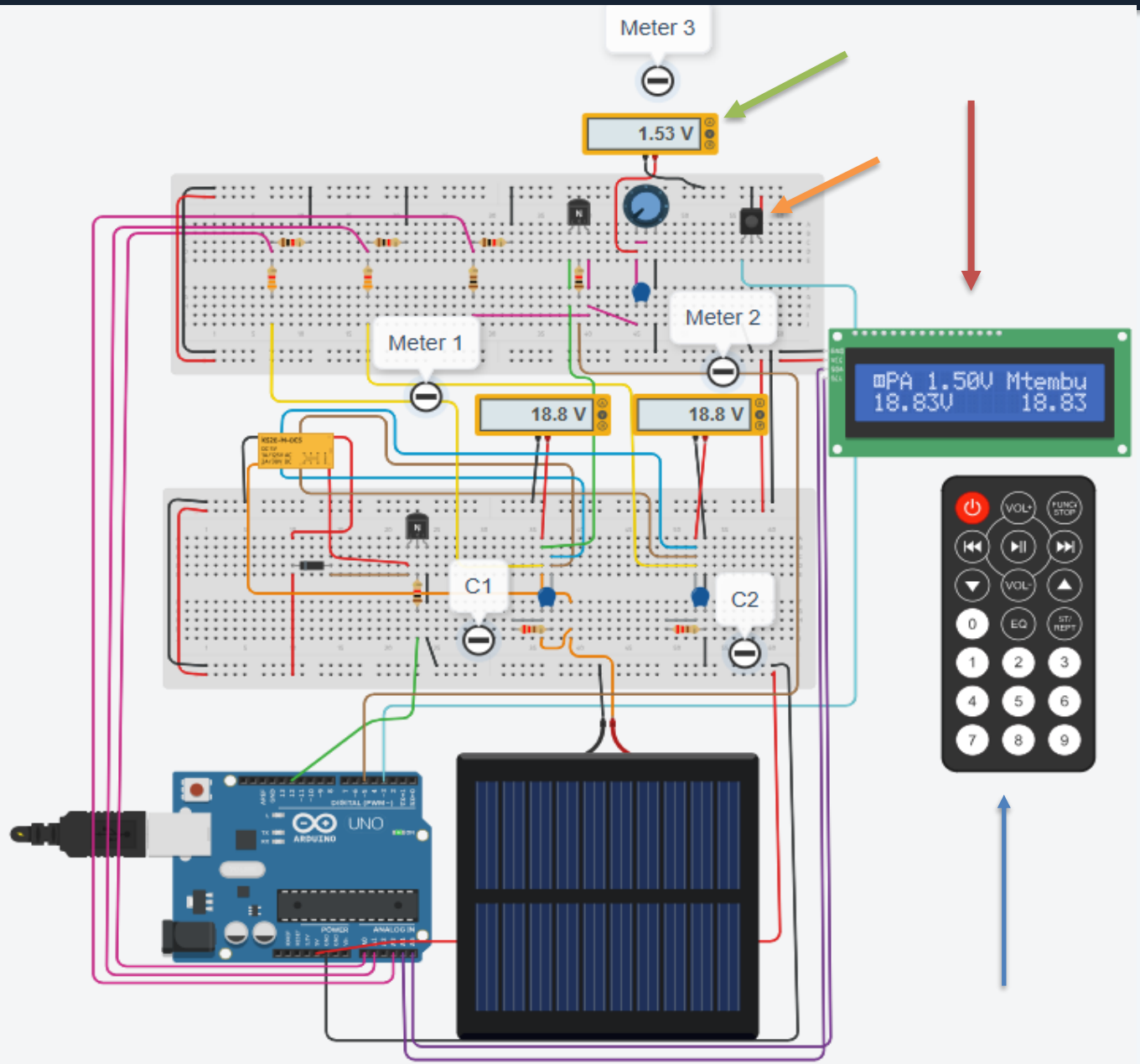
Phase 5 — Closed-Loop PWM Regulation

- Purpose: automatically regulate output voltage to the selected target.
- NPN transistor is controlled by PWM on Arduino Pin 5.
- A3 measures the regulated output and provides feedback.
- An Exponential Moving Average (EMA) smooths rapid measurement fluctuations.
- Arduino compares the filtered voltage with the target.
- PWM is adjusted from 0–255 to control transistor conduction.



Phase 6 — Final System Merge

- All tested subsystems are combined into the final embedded system.
- IR remote sets the target voltage.
- Voltage sensing provides feedback for the controller.
- Relay selects series or parallel topology when required.
- NPN transistor regulates the output using PWM and A3 feedback.
- LCD displays voltages, topology icons and the surname “Mtembu”.



Switching Condition & Control Logic

Switch to SERIES

If Stored Voltage < Target Voltage + 0.40 V

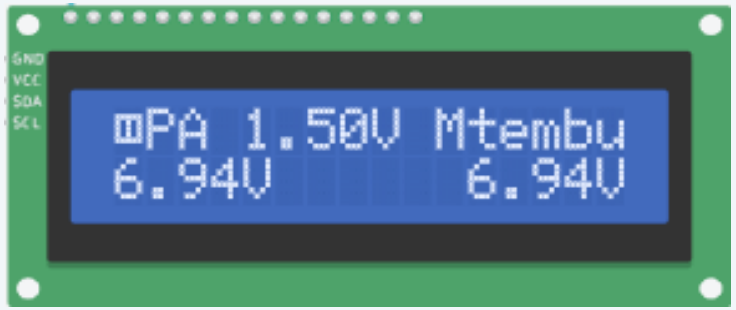
Series operation increases available supply voltage and provides regulator headroom.

Remain in PARALLEL

If Stored Voltage ≥ Target Voltage + 0.40 V

Parallel operation is retained because the required output can already be achieved.





Measure stored voltage → **Compare with target + 0.40 V** → **Select topology** → **Regulate output**

System Verification & Demonstration

1. LCD State

Custom series/parallel icons indicate the controller's selected operating state.

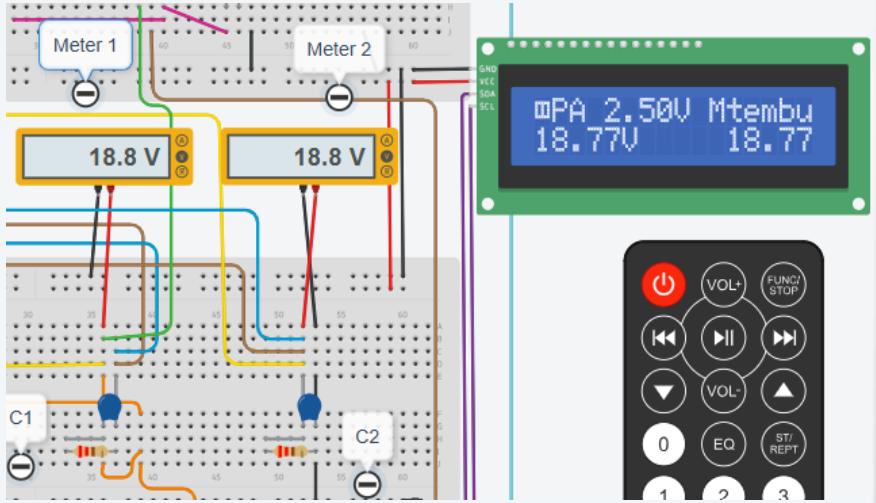
2. LCD Voltages

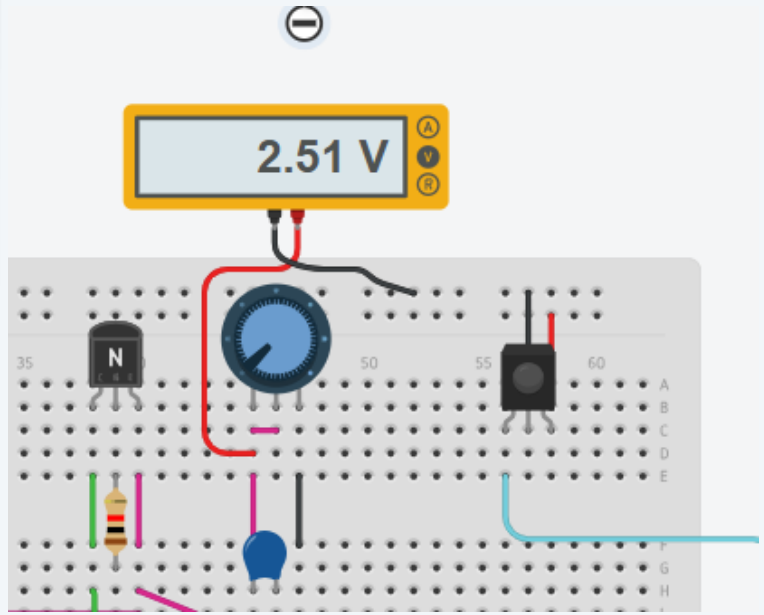
Voltage-divider readings show the measured capacitor and output voltages.

3. Physical Voltmeter

Independent electrical measurement verifies the displayed output voltage.

- Set a target voltage using the IR remote.
- LCD displays Input = 2.50V.
- Arduino stores the target and evaluates the available voltage.
- Relay changes topology when the switching condition is met.
- NPN transistor regulates the output through PWM.
- LCD and physical voltmeter confirm the resulting output.





Results & System Performance

Energy Storage

Capacitors successfully used as the battery substitute.

Dynamic Switching

Relay successfully changes between series and parallel.

Measurement

Voltage dividers, LCD readings and voltmeter provide verification.

User Control

IR remote selects the required output target.

Regulation

NPN transistor, PWM and A3 feedback form the closed loop.

Integration

Storage, sensing, switching, input and regulation operate together.

Scope, Limitations

Scope

Demonstrates capacitor-based storage, dynamic topology control and closed-loop voltage regulation for a small solar-system prototype.

Limitations

Operation is constrained by component voltage/current ratings and measurement range.

- Capacitors are suitable for demonstrating fast electrical energy storage and topology control.
- The prototype is not intended to directly replace large battery-storage systems.
- Future work should focus on higher energy capacity and improved solar-energy extraction.

Conclusion

- A solar-powered energy-storage system using capacitors instead of conventional batteries was developed.
- The capacitor bank can dynamically operate in series or parallel.
- Voltage-divider circuits provide safe system monitoring.
- An IR remote allows the user to select the required output voltage.
- An NPN transistor provides PWM-based voltage control.
- A3 feedback creates a closed-loop regulation system.
- EMA filtering improves feedback stability.
- LCD indicators and physical voltmeter measurements provide system verification.

Final System Concept

Solar Energy → Capacitor Storage → Dynamic Topology → Closed-Loop Regulation → Controlled Output